Lecture 21 Scribe

L21_L22.pdf
PDF

Generate a lecture scribe intended to serve as exam-oriented reference material for Lecture 21(ppt uploaded). Use only the provided context (lecture slides and relevant textbook content). While writing the scribe, reason step by step through definitions, assumptions, proofs, derivations, and example solutions exactly as presented in the context. Make the reasoning explicit so the document can be used directly for revision. Include: Definitions and notation Assumptions and conditions Theorem or result statements Step-by-step proofs or derivations when present Worked examples with intermediate reasoning Logical structure reflecting the lecture flow Maintain a formal academic tone appropriate for exam study. Do not invent content, add intuition not discussed in lecture, or include material beyond the provided context. Ensure all reasoning is grounded strictly in the lecture and textbook material. Finally, please ensure that each and every thing in the ppt , all concepts, all slides , all pages are covered.

Reading documents

Here is your **complete exam-oriented lecture scribe for Lecture 21 (Inequalities & Tail Bounds)**, strictly based on the provided slides. All definitions, derivations, proofs, and examples are covered in the exact logical flow of the lecture.

📘 Lecture 21: Inequalities and Tail Bounds

(From)

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1. Tails of a Random Variable

Definition

The **tail** of a random variable X is:

$$P(X > x)$$

This represents the probability that X takes **large/extreme values**.

Motivation: Why do we care about tails?

From slides:

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer

Key Idea

- Mean → average behavior (easy to compute)
- Tail → rare extreme events (hard to compute)

Difficulty

- Tail computation requires **summing many small probabilities**

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2. Tail Examples

Example 1: Load Balancing (Binomial Tail)

Problem

Distribute n jobs among n servers randomly.
Find:

$$P(X \geq k)$$

Model

$$X \sim \text{Binomial}(n, p)$$

Exact Tail

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Interpretation

- Most servers → few jobs
- Rarely → one server gets many jobs
- This is a **tail event (rare overload)**

Example 2: Poisson Tail

Problem

Jobs arrive with rate λ.
Find:

$$P(X \geq k)$$

Model

$$X \sim \text{Poisson}(\lambda)$$

Exact Tail

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Interpretation

- Most times → moderate arrivals
- Rarely → burst arrivals
- Tail event = **sudden spike**

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3. Why Tail Bounds?

Problem

Exact computation:

- Binomial → long sum
- Poisson → infinite sum

Idea

Instead of exact value:

$$P(X \geq k) \leq \text{simple upper bound}$$

Advantages

- Easier to compute
- Still guarantees correctness

Key Principle

👉 Approximate safely instead of computing exactly

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4. Running Example (Used Throughout)

Experiment

Flip a fair coin n times

Random Variable

$$X = \text{number of heads}$$

$$X \sim \text{Binomial}(n, \tfrac{1}{2})$$

Mean

$$E[X] = \frac{n}{2}$$

Question

$$P\left(X \geq \frac{3n}{4}\right)$$

This is a **tail event (many heads)**.

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5. Markov's Inequality

Statement

For non-negative random variable X:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Assumptions

- $X \geq 0$
- Finite expectation

Key Idea

- Uses only the mean
- Large values are rare

Proof (Step-by-step)

Start from expectation:

$$E[X] = \sum_{x=0}^{\infty} x \cdot p_X(x)$$

Restrict to $x \geq a$:

$$E[X] \geq \sum_{x=a}^{\infty} x \cdot p_X(x)$$

Replace x by a (since $x \geq a$):

$$\geq \sum_{x=a}^{\infty} a \cdot p_X(x)$$

$$= a \cdot P(X \geq a)$$

Thus:

$$E[X] \geq a \cdot P(X \geq a)$$

$$\Rightarrow P(X \geq a) \leq \frac{E[X]}{a}$$

Running Example Application

$$X \sim \text{Binomial}(n, 1/2), \quad E[X] = n/2$$

$$P\left(X \geq \frac{3n}{4}\right) \leq \frac{n/2}{3n/4} = \frac{2}{3}$$

Observation

- Bound does **not depend on** n
- Hence **very weak (loose bound)**

Real-Life Example

Income Distribution

- Average income = 10,000
- Find $P(X \geq 50,000)$

Using Markov:

$$P(X \geq 50000) \leq \frac{10000}{50000} = 0.2$$

👉 At most 20% people earn ≥ 50,000

Solved Example 1 (Exam Grades)

Given

- Mean = 70
- Threshold = 90

Steps

1. $X \geq 0$, $E[X] = 70$
2. Apply Markov:

$$P(X \geq 90) \leq \frac{70}{90}$$

3. Simplify:

$$= \frac{7}{9} \approx 0.778$$

Solved Example 2 (Server Load)

Given

- Mean = 5000
- Threshold = 20000

Steps

$$P(X \geq 20000) \leq \frac{5000}{20000} = \frac{1}{4} = 0.25$$

Homework Insight

- Markov bounds **right tail only**
- Cannot directly handle $P(X < a)$

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6. Chebyshev's Inequality

Key Idea

- Uses **mean + variance**
- Controls deviation from mean

Statement

$$P(|X-\mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Proof Idea

Apply Markov on:

$$(X-\mu)^2$$

$$P(|X-\mu| \geq a) = P((X-\mu)^2 \geq a^2)$$

Apply Markov:

$$\leq \frac{E[(X-\mu)^2]}{a^2}$$

$$= \frac{\mathrm{Var}(X)}{a^2}$$

Running Example

$$X \sim \mathrm{Binomial}(n, 1/2)$$

$$\mu = \frac{n}{2}, \quad \mathrm{Var}(X) = \frac{n}{4}$$

We want:

$$P\left(X \geq \frac{3n}{4}\right)$$

Rewrite:

$$P\left(X - \frac{n}{2} \geq \frac{n}{4}\right)$$

$$\leq P\left(|X - \frac{n}{2}| \geq \frac{n}{4}\right)$$

Apply Chebyshev:

$$\leq \frac{n/4}{(n/4)^2} = \frac{4}{n}$$

Insight

- Bound improves as $n \uparrow$
- Much better than Markov

Real-Life Example (Exam Scores)

Given

- Mean = 70
- Std dev = 10

Question

$$P(|X - 70| \geq 30)$$

Solution

$$\leq \frac{10^2}{30^2} = \frac{100}{900} = \frac{1}{9}$$

Solved Example 1

Given

- Mean = 70
- Std dev = 5 → variance = 25
- Threshold = 90

Steps

$$P(X \geq 90) \leq P(|X - 70| \geq 20)$$

$$\leq \frac{25}{400} = \frac{1}{16} = 0.0625$$

👉 Much tighter than Markov (0.778)

Solved Example 2 (Latency)

Given

- Mean = 50
- Variance = 16
- Range: [30, 70]

Convert

$$|X - 50| \geq 20$$

Apply Chebyshev:

$$\leq \frac{16}{400} = \frac{1}{25} = 0.04$$

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7. Chernoff Bound

Motivation

Progression:

- Markov → uses X
- Chebyshev → uses $(X - \mu)^2$
- Chernoff → uses e^{tX}

Key Insight

- Exponential grows fast
- Amplifies large values
- Gives tighter bounds

Core Transformation

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

Apply Markov

$$P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}$$

Final Form

$$P(X \geq a) \leq \min_{t>0} \frac{E[e^{tX}]}{e^{ta}}$$

Key Properties

- Works for all $t > 0$
- Choose best t → tightest bound
- Much stronger than previous bounds
- Gives **exponential decay**

Lower Tail Version

Goal:

$$P(X \leq a)$$

Use:

$$t < 0$$

$$P(X \leq a) \leq \min_{t<0} \frac{E[e^{tX}]}{e^{ta}}$$

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8. Chernoff Bound for Poisson

Step 1: Compute MGF

$$E[e^{tX}] = \sum_{i=0}^{\infty} e^{ti} e^{-\lambda} \frac{\lambda^i}{i!}$$

$$= e^{-\lambda} \sum_{i=0}^{\infty} \frac{(\lambda e^t)^i}{i!}$$

$$= e^{-\lambda} e^{\lambda e^t}$$

$$= e^{\lambda(e^t - 1)}$$

Final Insight

- Plug into Chernoff bound
- Optimize over t

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9. Summary of Bounds

Inequality	Uses	Bound Type
Markov	Mean	Weak
Chebyshev	Mean + Variance	Better
Chernoff	Exponential	Strong (tight)

🔥 Final Takeaways (Exam Ready)

- Tail = $P(X > x)$ → rare events
- Exact computation = difficult
- Use bounds instead

Hierarchy

$$\text{Chernoff} > \text{Chebyshev} > \text{Markov}$$

- Markov → simple but loose
- Chebyshev → depends on variance
- Chernoff → exponential decay (best)

If you want, I can convert this into **1-page revision notes / cheat sheet** or give **expected exam questions from this lecture**.